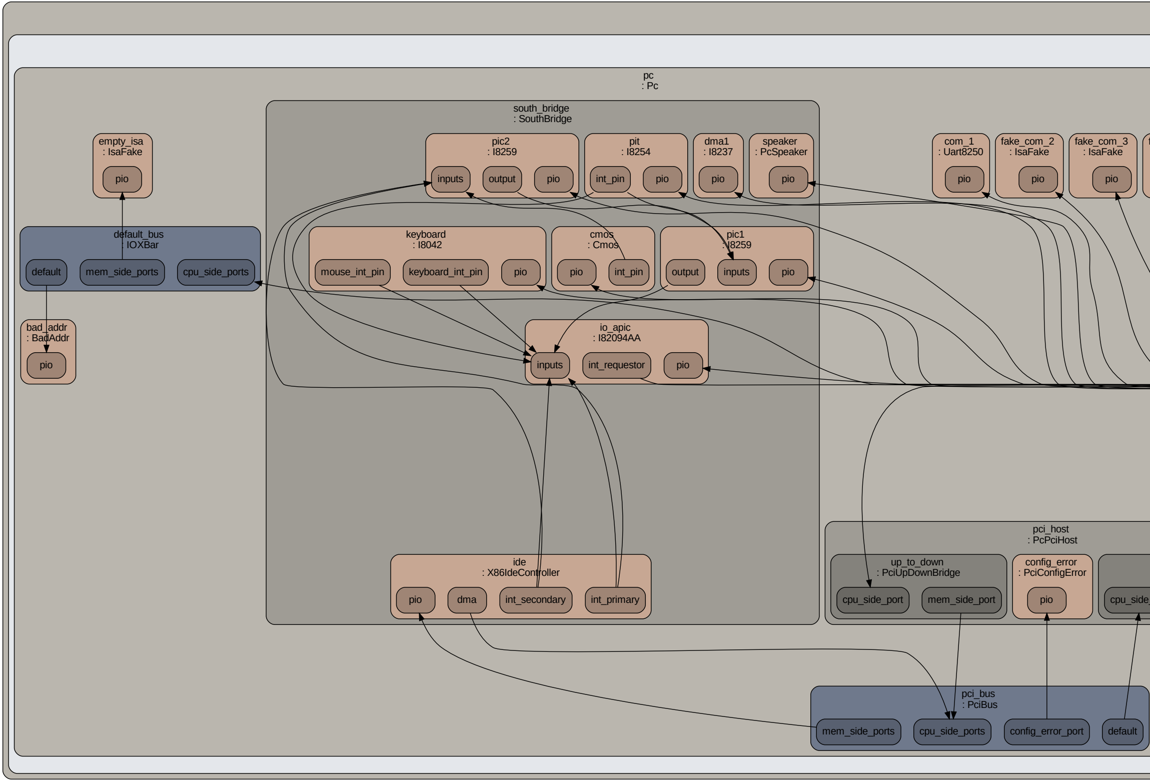
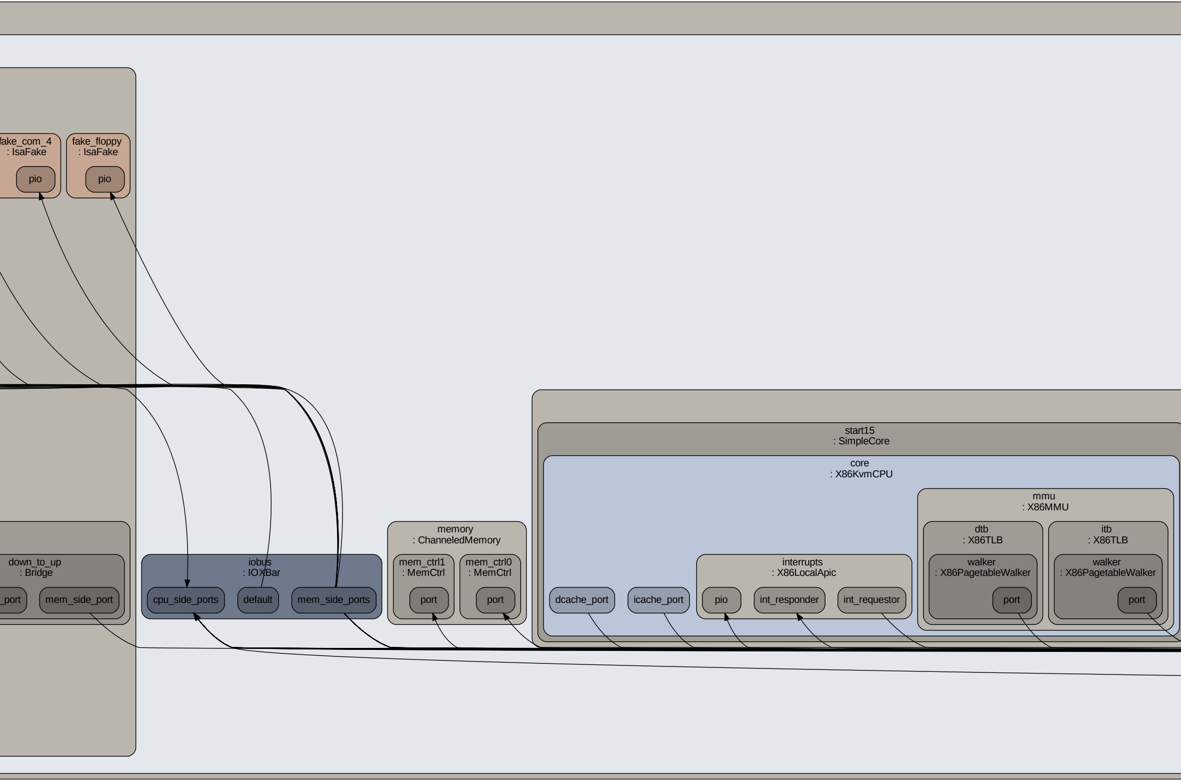
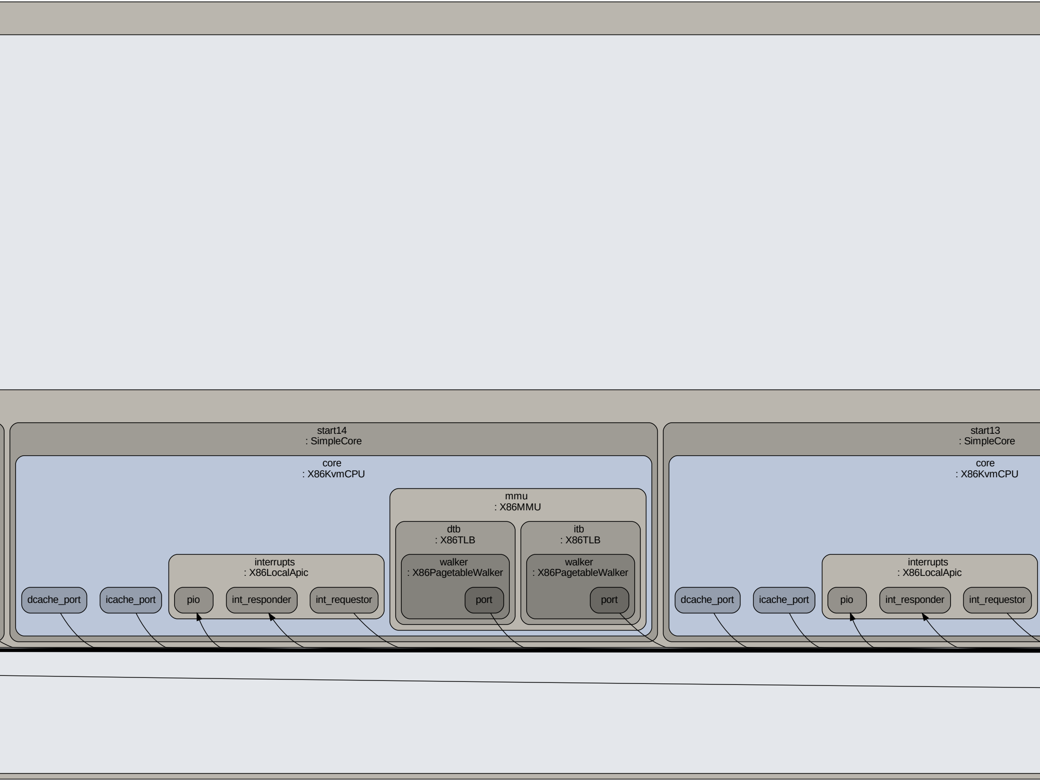
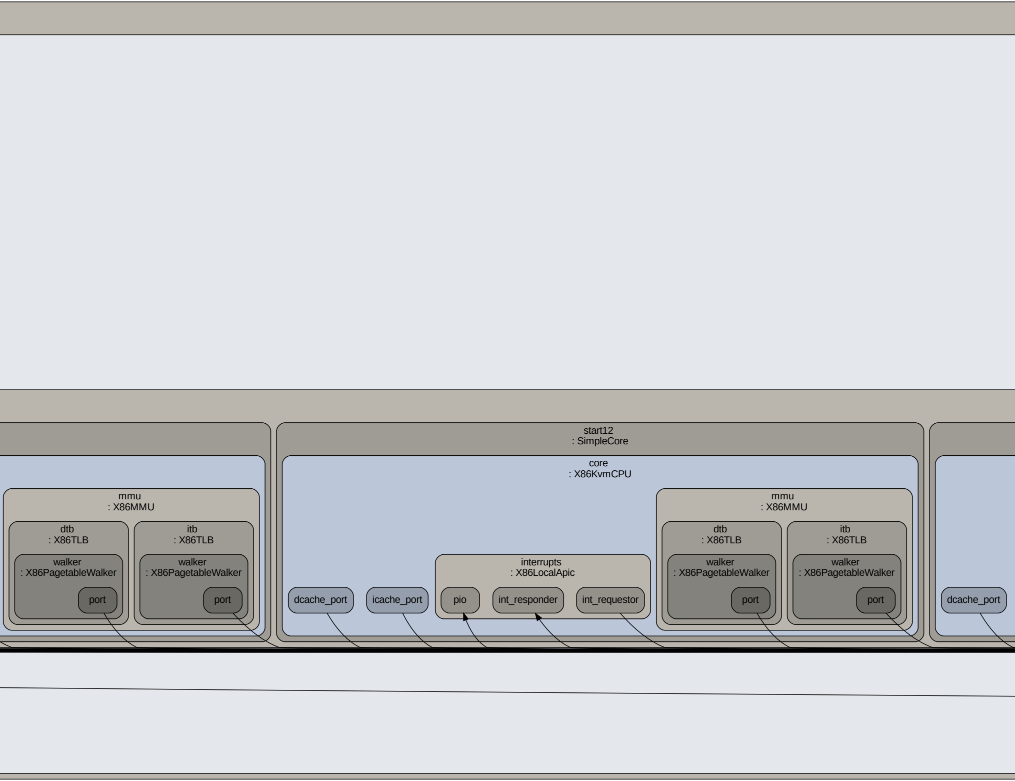
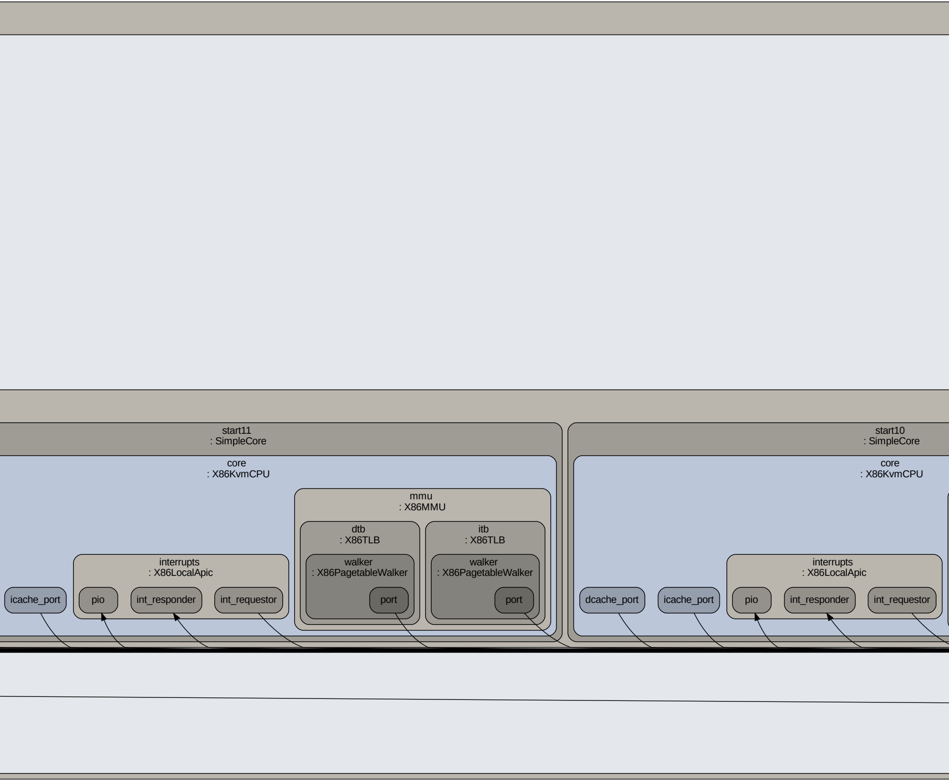

pc
: Pc
south_bridge
: SouthBridge
empty_isa
: IsaFake
pio
pic2
: I8259
inputs
output
pio
pit
: I8254
int_pin
pio
dma1
: I8237
pio
speaker
: PcSpeaker
pio
com_1
: Uart8250
pio
fake_com_2
: IsaFake
pio
fake_com_3
: IsaFake
pio
default_bus
: IOXBar
default
mem_side_ports
cpu_side_ports
keyboard
: I8042
mouse_int_pin
keyboard_int_pin
pio
cmos
: Cmos
pio
int_pin
pic1
: I8259
output
inputs
pio
bad_addr
: BadAddr
pio
io_apic
: I82094AA
inputs
int_requestor
pio
ide
: X86IdeController
pio
dma
int_secondary
int_primary
pci_host
: PcPciHost
up_to_down
: PciUpDownBridge
cpu_side_port
mem_side_port
config_error
: PciConfigError
pio
cpu_side_
pci_bus
: PciBus
mem_side_ports
cpu_side_ports
config_error_port
default

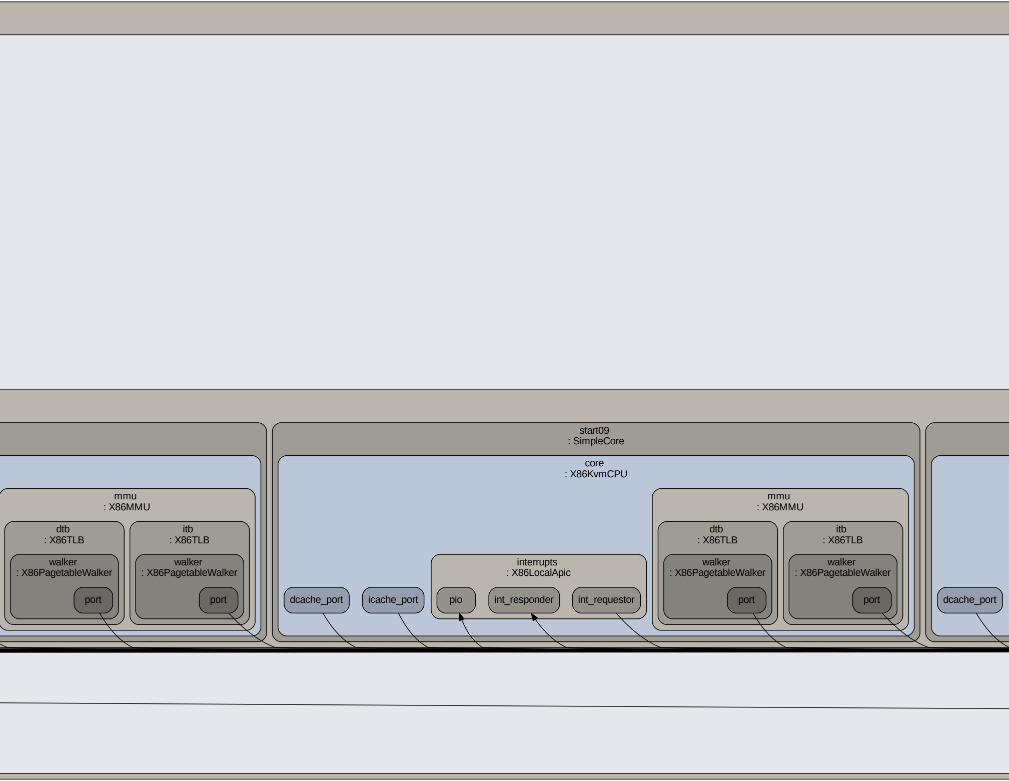

fake_com_4
: IsaFake
pio
fake_floppy
: IsaFake
pio
down_to_up
: Bridge
_port
mem_side_port
iobus
: IOXBar
cpu_side_ports
default
mem_side_ports
memory
: ChanneledMemory
mem_ctrl1
: MemCtrl
port
mem_ctrl0
: MemCtrl
port
start15
: SimpleCore
core
: X86KvmCPU
dcache_port
icache_port
interrupts
: X86LocalApic
pio
int_responder
int_requestor
mmu
: X86MMU
dtb
: X86TLB
walker
: X86PagetableWalker
port
itb
: X86TLB
walker
: X86PagetableWalker
port

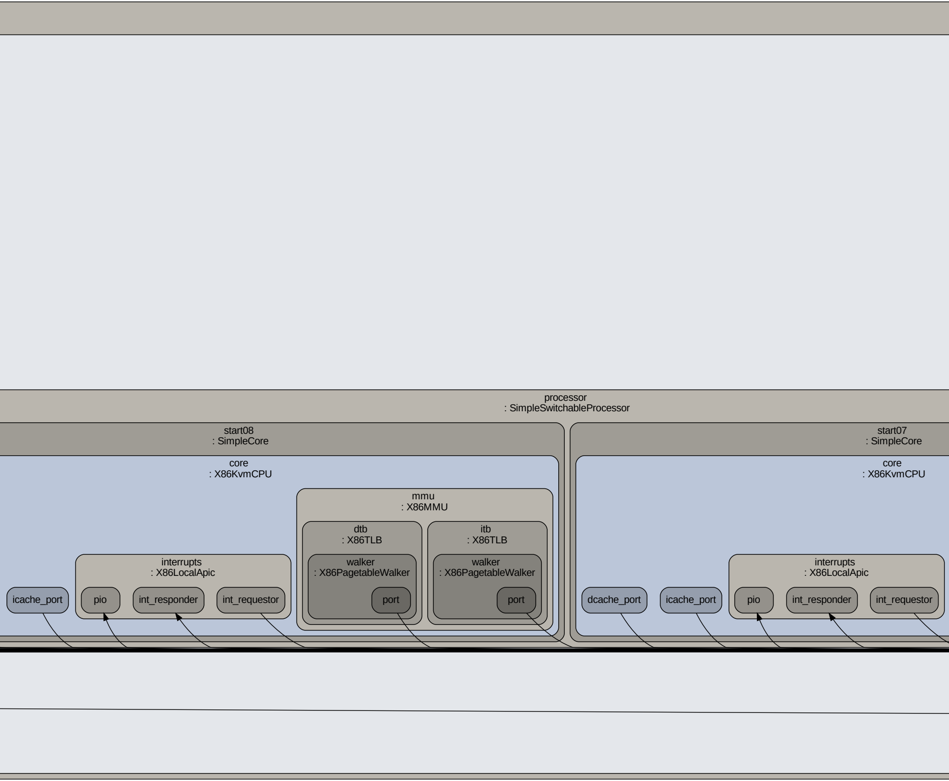

start14
: SimpleCore
core
: X86KvmCPU
mmu
: X86MMU
dtb
: X86TLB
itb
: X86TLB
walker
: X86PagetableWalker
walker
: X86PagetableWalker
port
port
interrupts
: X86LocalApic
dcache_port
icache_port
pio
int_responder
int_requestor
start13
: SimpleCore
core
: X86KvmCPU
interrupts
: X86LocalApic
dcache_port
icache_port
pio
int_responder
int_requestor

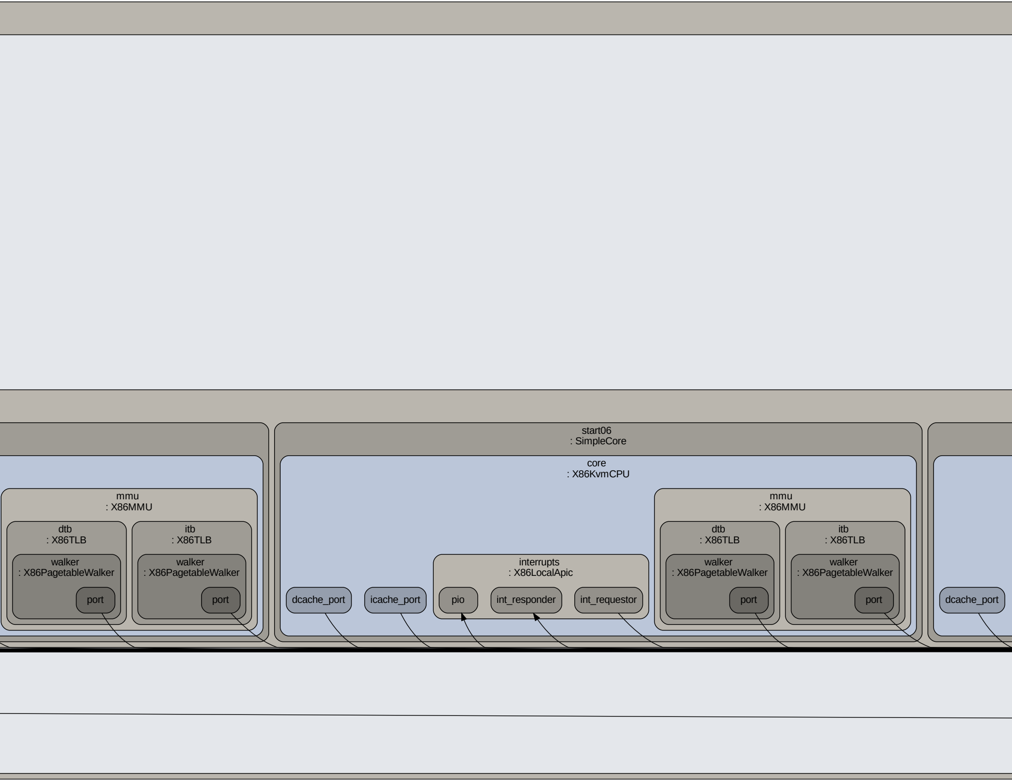


mmu
: X86MMU
dtb
: X86TLB
walker
: X86PagetableWalker
port
itb
: X86TLB
walker
: X86PagetableWalker
port
start12
: SimpleCore
core
: X86KvmCPU
dcache_port
icache_port
interrupts
: X86LocalApic
pio
int_responder
int_requestor
mmu
: X86MMU
dtb
: X86TLB
walker
: X86PagetableWalker
port
itb
: X86TLB
walker
: X86PagetableWalker
port
dcache_port

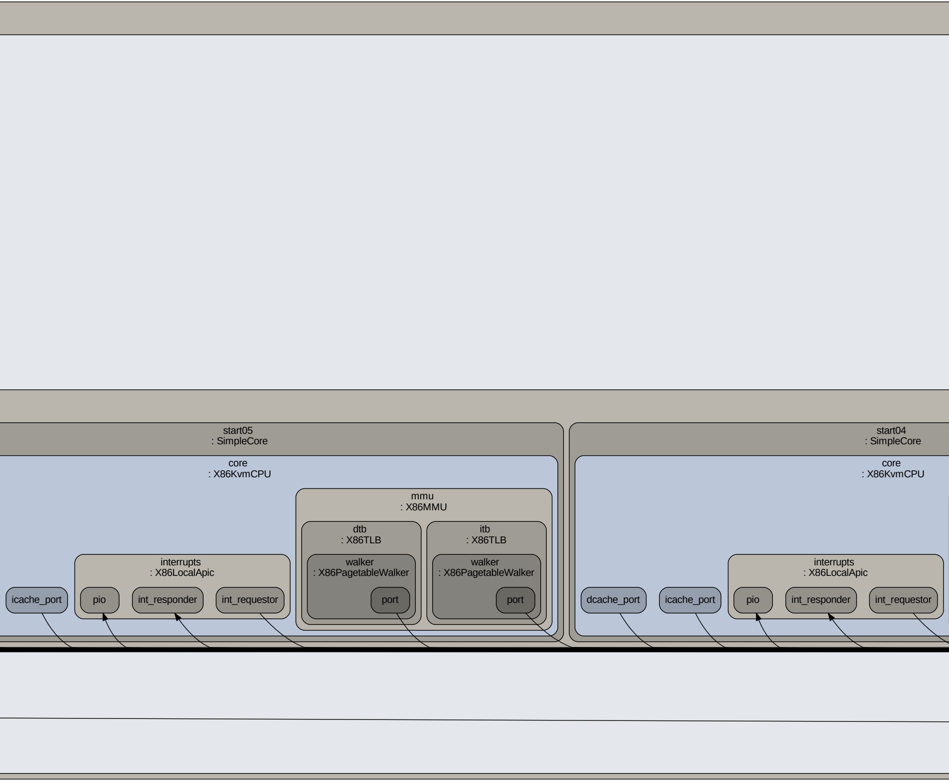

start11
: SimpleCore
core
: X86KvmCPU
mmu
: X86MMU
dtb
: X86TLB
itb
: X86TLB
walker
: X86PagetableWalker
port
walker
: X86PagetableWalker
port
icache_port
interrupts
: X86LocalApic
pio
int_responder
int_requestor
start10
: SimpleCore
core
: X86KvmCPU
dcache_port
icache_port
interrupts
: X86LocalApic
pio
int_responder
int_requestor

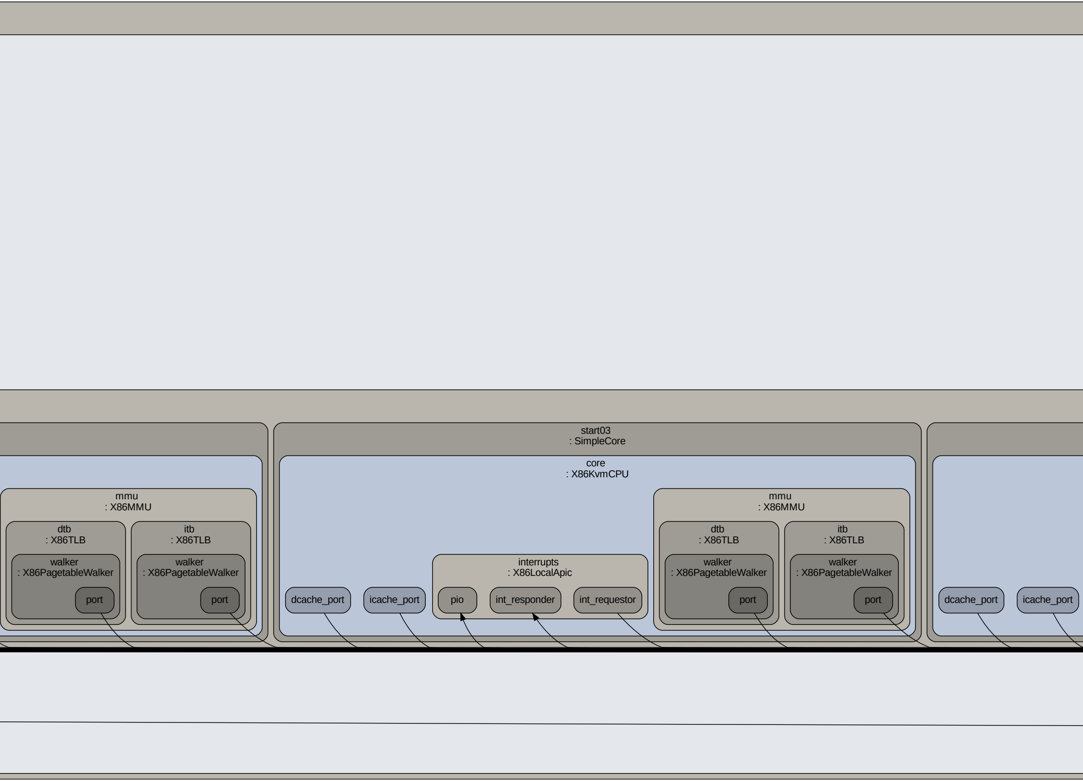

mmu
: X86MMU
dtb
: X86TLB
walker
: X86PagetableWalker
port
itb
: X86TLB
walker
: X86PagetableWalker
port
start09
: SimpleCore
core
: X86KvmCPU
dcache_port
icache_port
interrupts
: X86LocalApic
pio
int_responder
int_requestor
mmu
: X86MMU
dtb
: X86TLB
walker
: X86PagetableWalker
port
itb
: X86TLB
walker
: X86PagetableWalker
port
dcache_port

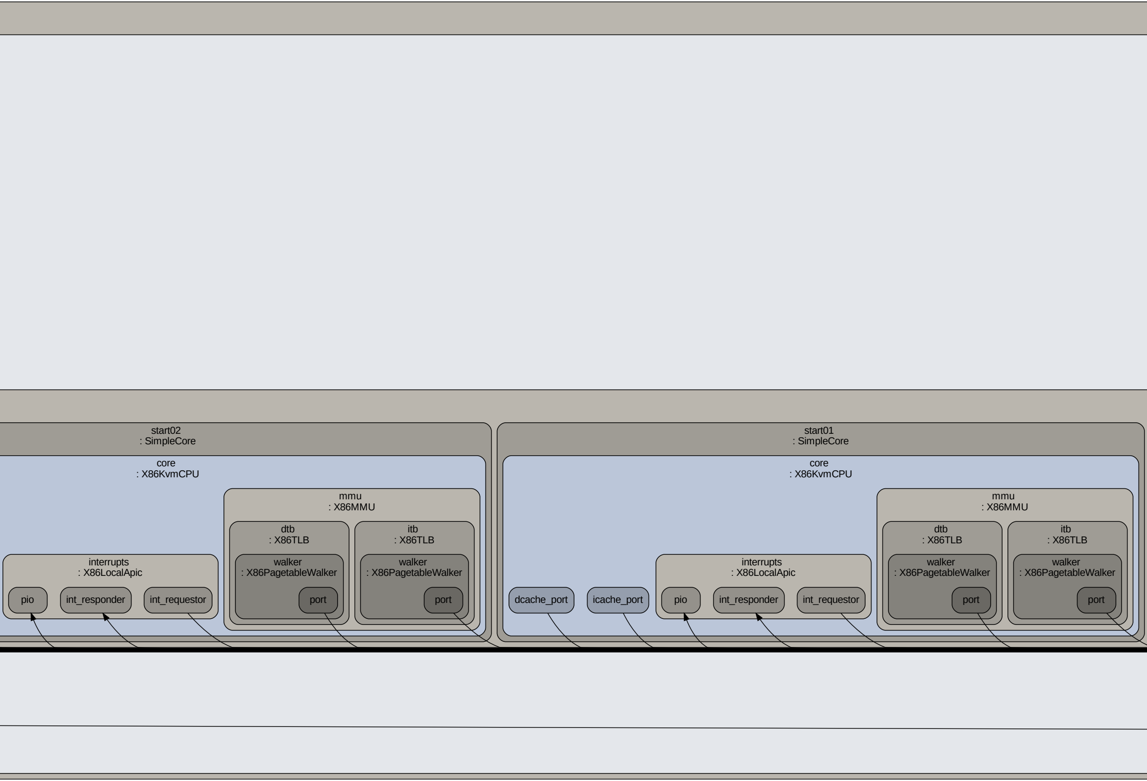

processor
: SimpleSwitchableProcessor
start08
: SimpleCore
core
: X86KvmCPU
mmu
: X86MMU
dtb
: X86TLB
walker
: X86PagetableWalker
port
itb
: X86TLB
walker
: X86PagetableWalker
port
interrupts
: X86LocalApic
icache_port
pio
int_responder
int_requestor
start07
: SimpleCore
core
: X86KvmCPU
dcache_port
icache_port
interrupts
: X86LocalApic
pio
int_responder
int_requestor

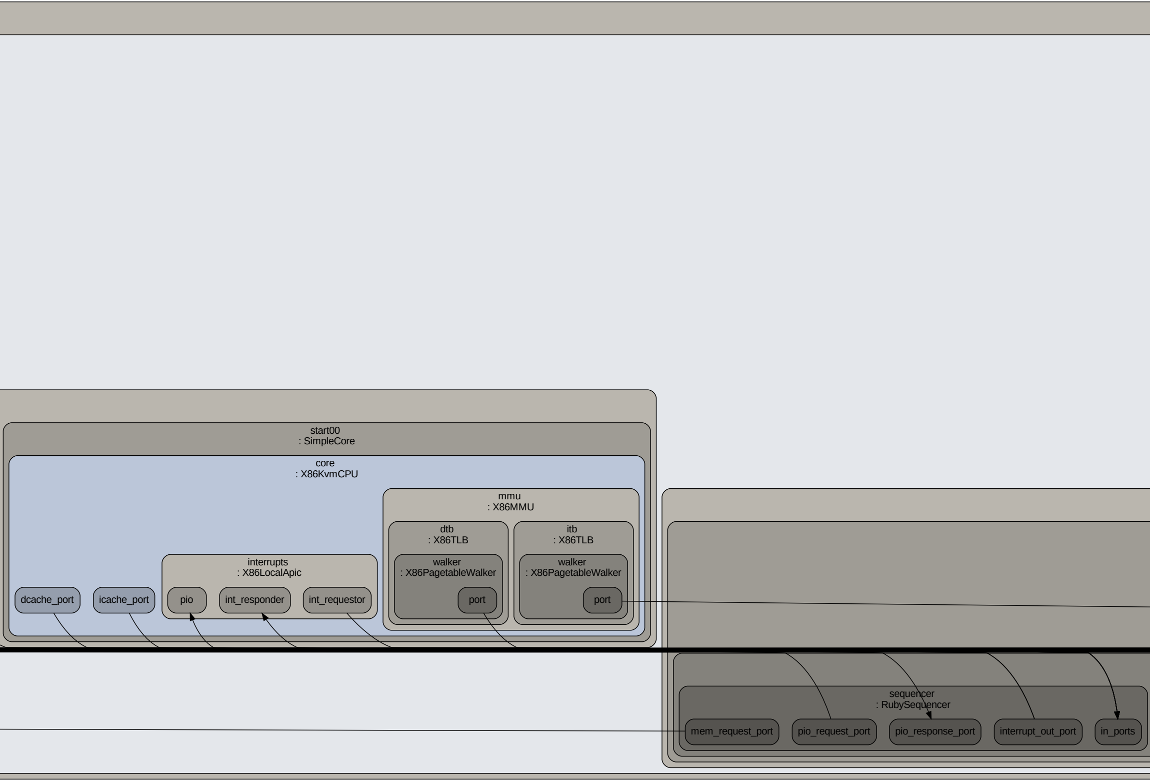


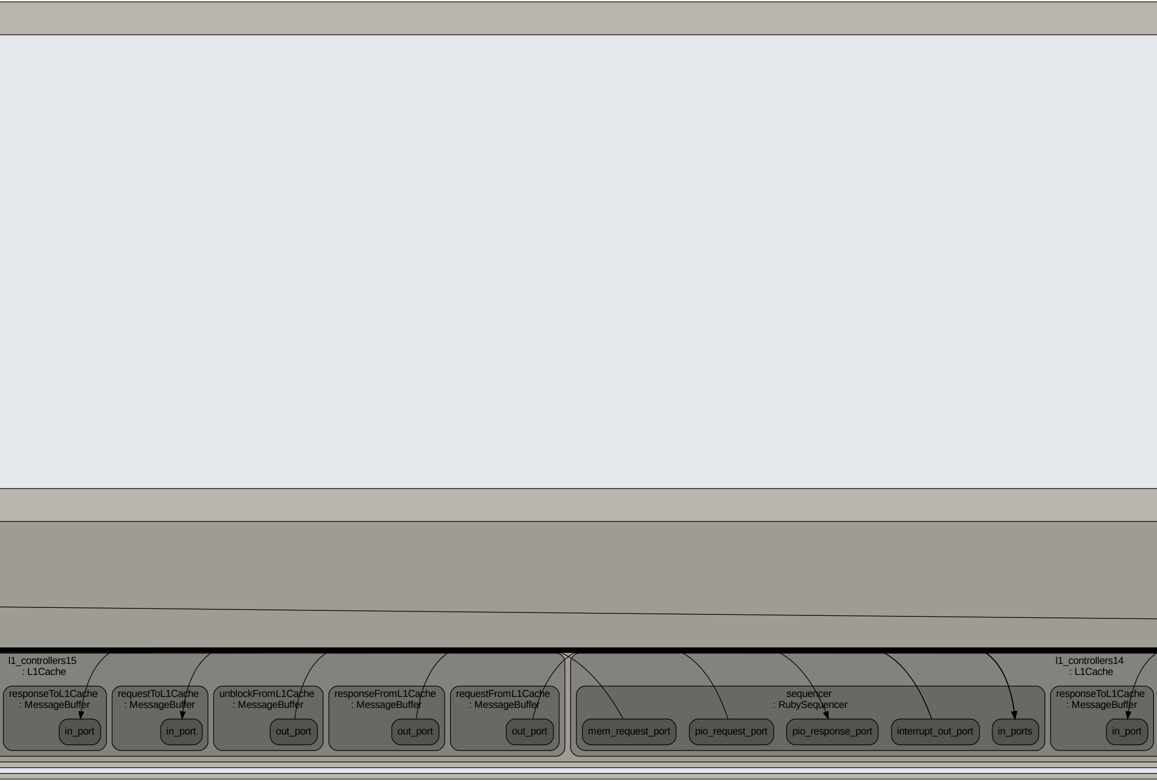
mmu
: X86MMU
dtb
: X86TLB
walker
: X86PagetableWalker
port
itb
: X86TLB
walker
: X86PagetableWalker
port
start06
: SimpleCore
core
: X86KvmCPU
dcache_port
icache_port
interrupts
: X86LocalApic
pio
int_responder
int_requestor
mmu
: X86MMU
dtb
: X86TLB
walker
: X86PagetableWalker
port
itb
: X86TLB
walker
: X86PagetableWalker
port
dcache_port

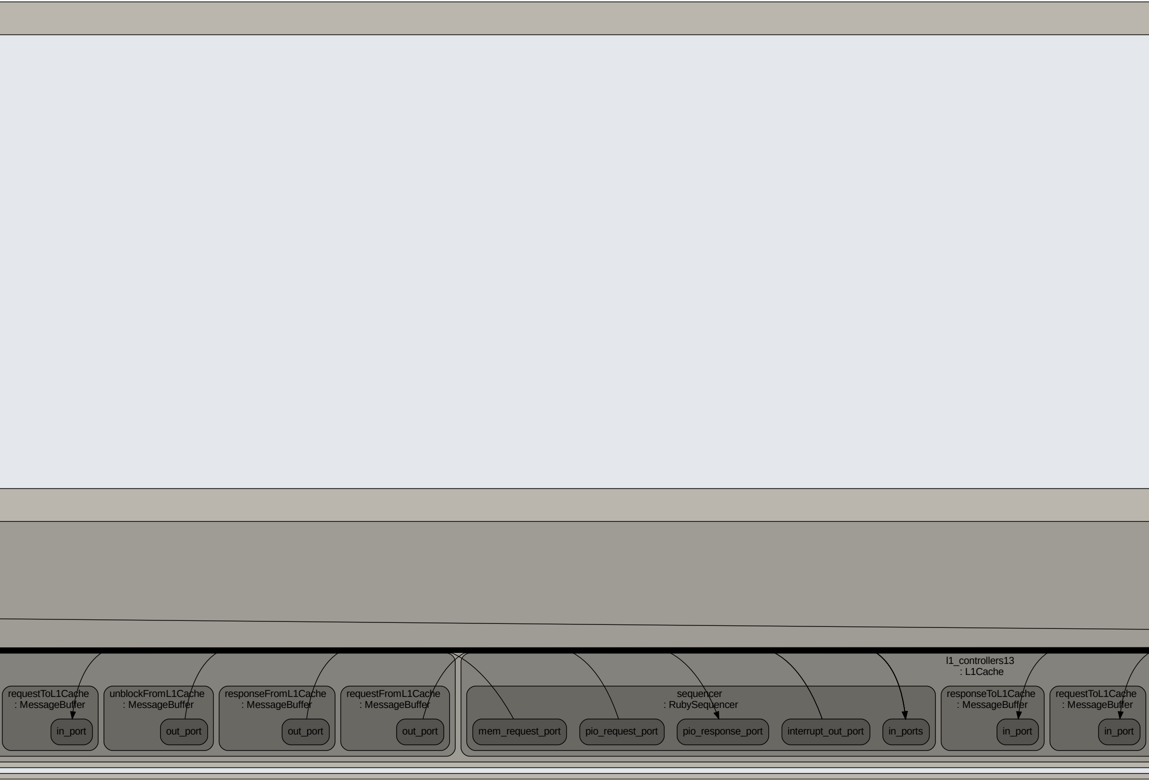

start05
: SimpleCore
core
: X86KvmCPU
mmu
: X86MMU
dtb
: X86TLB
walker
: X86PagetableWalker
port
itb
: X86TLB
walker
: X86PagetableWalker
port
interrupts
: X86LocalApic
pio
int_responder
int_requestor
icache_port
start04
: SimpleCore
core
: X86KvmCPU
interrupts
: X86LocalApic
dcache_port
icache_port
pio
int_responder
int_requestor

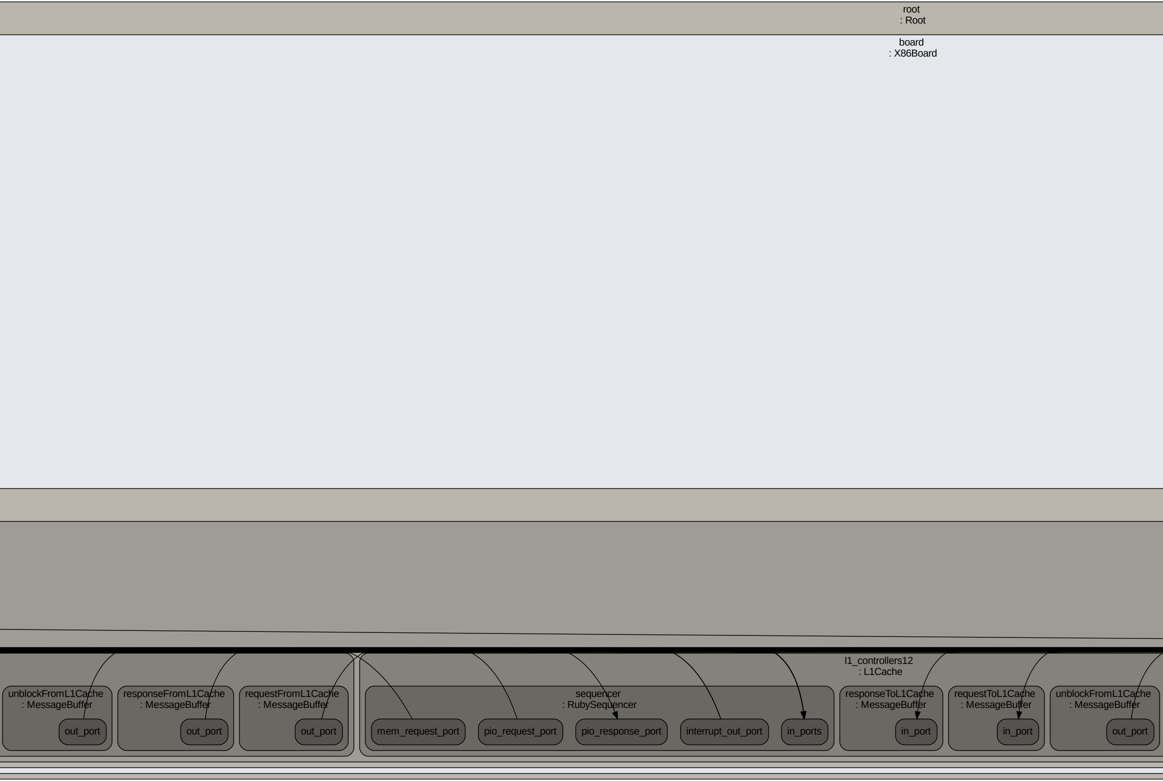

mmu
: X86MMU
dtb
: X86TLB
walker
: X86PagetableWalker
port
itb
: X86TLB
walker
: X86PagetableWalker
port
start03
: SimpleCore
core
: X86KvmCPU
dcache_port
icache_port
interrupts
: X86LocalApic
pio
int_responder
int_requestor
mmu
: X86MMU
dtb
: X86TLB
walker
: X86PagetableWalker
port
itb
: X86TLB
walker
: X86PagetableWalker
port
dcache_port
icache_port

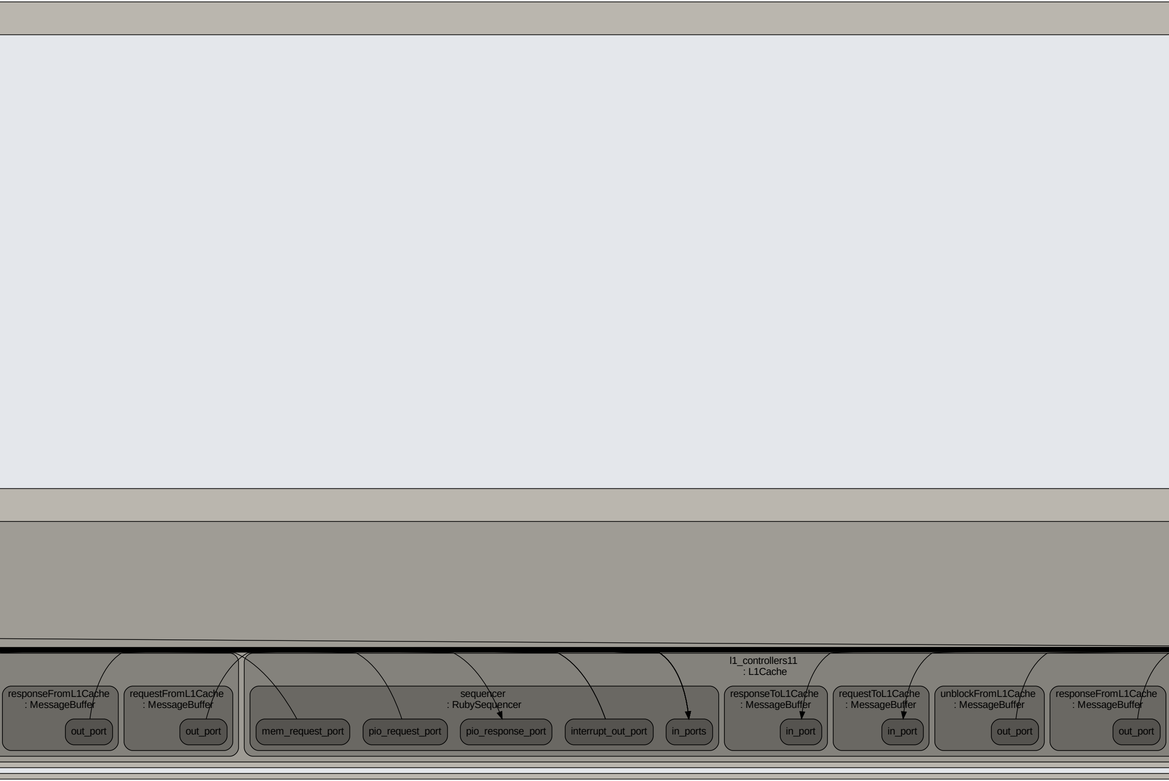

start02
: SimpleCore
core
: X86KvmCPU
mmu
: X86MMU
dtb
: X86TLB
itb
: X86TLB
walker
: X86PagetableWalker
walker
: X86PagetableWalker
interrupts
: X86LocalApic
pio
int_responder
int_requestor
port
port
start01
: SimpleCore
core
: X86KvmCPU
mmu
: X86MMU
dtb
: X86TLB
itb
: X86TLB
walker
: X86PagetableWalker
walker
: X86PagetableWalker
interrupts
: X86LocalApic
dcache_port
icache_port
pio
int_responder
int_requestor
port
port

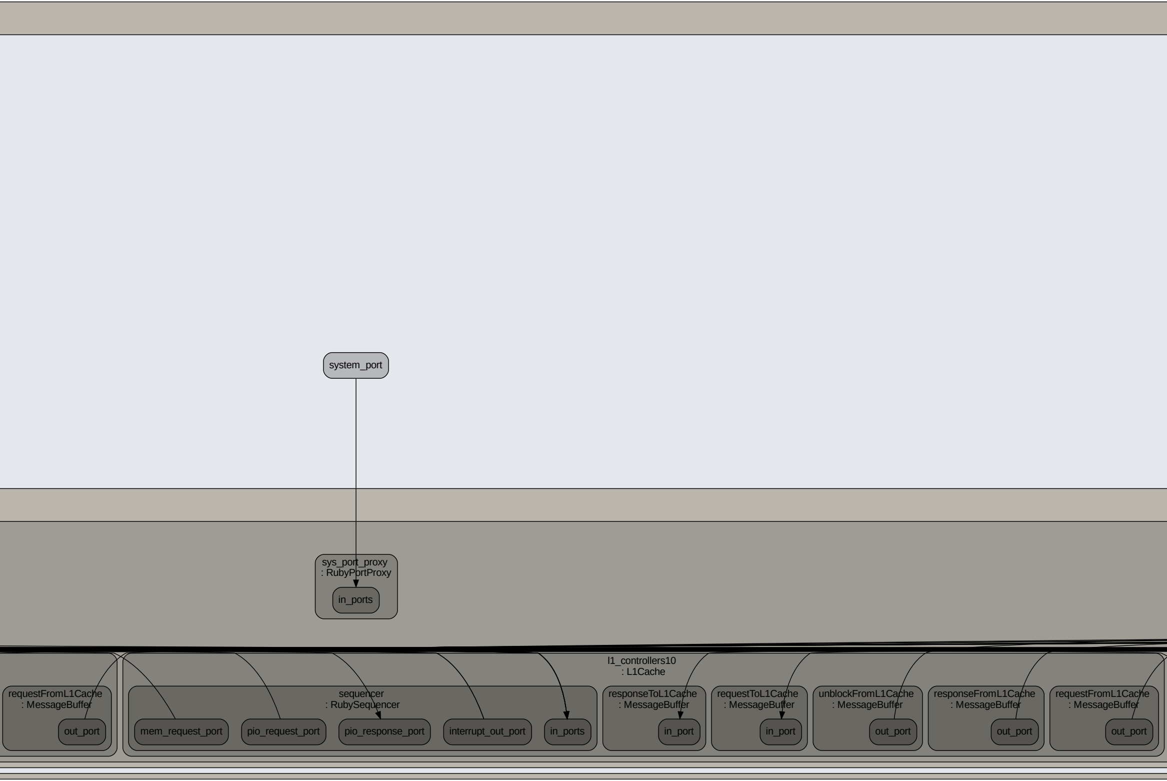

start00
: SimpleCore
core
: X86KvmCPU
mmu
: X86MMU
dtb
: X86TLB
itb
: X86TLB
walker
: X86PagetableWalker
walker
: X86PagetableWalker
port
port
interrupts
: X86LocalApic
dcache_port
icache_port
pio
int_responder
int_requestor
sequencer
: RubySequencer
mem_request_port
pio_request_port
pio_response_port
interrupt_out_port
in_ports

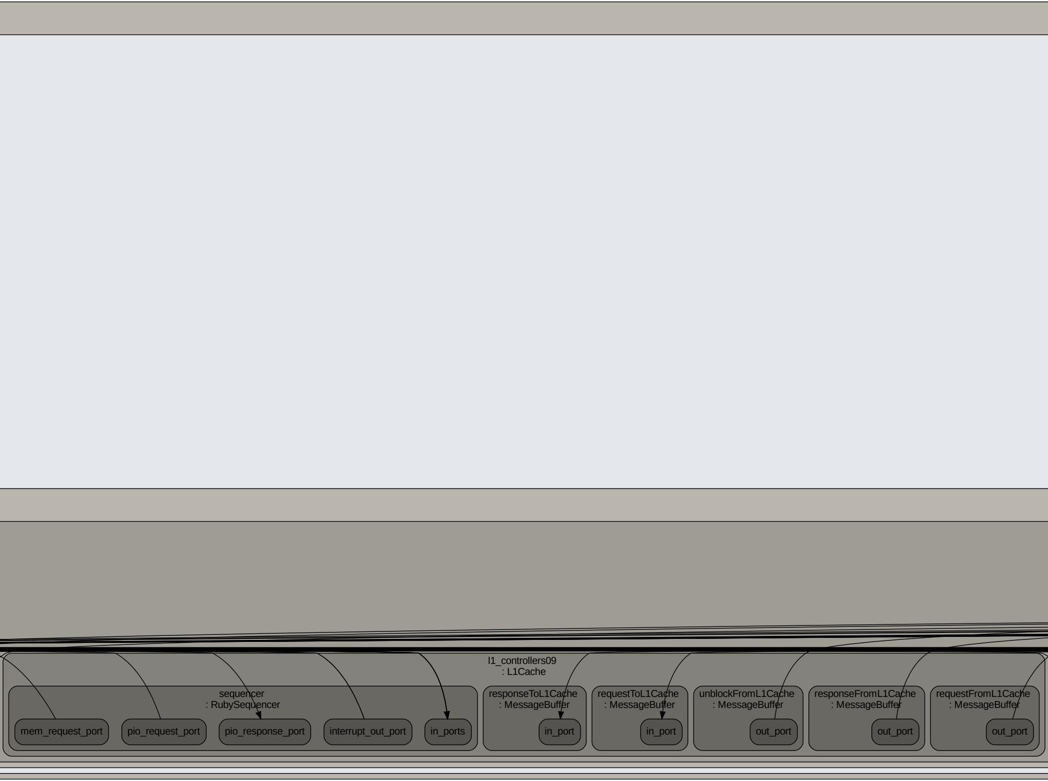

l1_controllers15
: L1Cache
responseToL1Cache
: MessageBuffer
in_port
requestToL1Cache
: MessageBuffer
in_port
unblockFromL1Cache
: MessageBuffer
out_port
responseFromL1Cache
: MessageBuffer
out_port
requestFromL1Cache
: MessageBuffer
out_port
sequencer
: RubySequencer
mem_request_port
pio_request_port
pio_response_port
interrupt_out_port
in_ports
l1_controllers14
: L1Cache
responseToL1Cache
: MessageBuffer
in_port

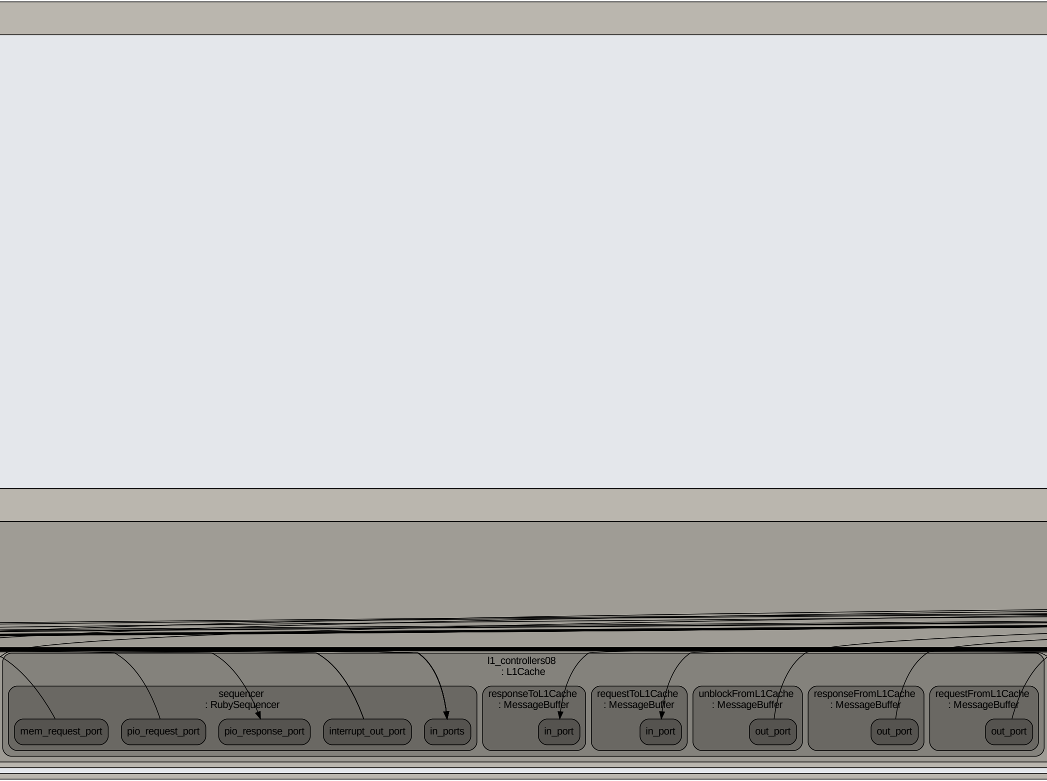

l1_controllers13
: L1Cache
requestToL1Cache
: MessageBuffer
in_port
unblockFromL1Cache
: MessageBuffer
out_port
responseFromL1Cache
: MessageBuffer
out_port
requestFromL1Cache
: MessageBuffer
out_port
sequencer
: RubySequencer
mem_request_port
pio_request_port
pio_response_port
interrupt_out_port
in_ports
responseToL1Cache
: MessageBuffer
in_port
requestToL1Cache
: MessageBuffer
in_port

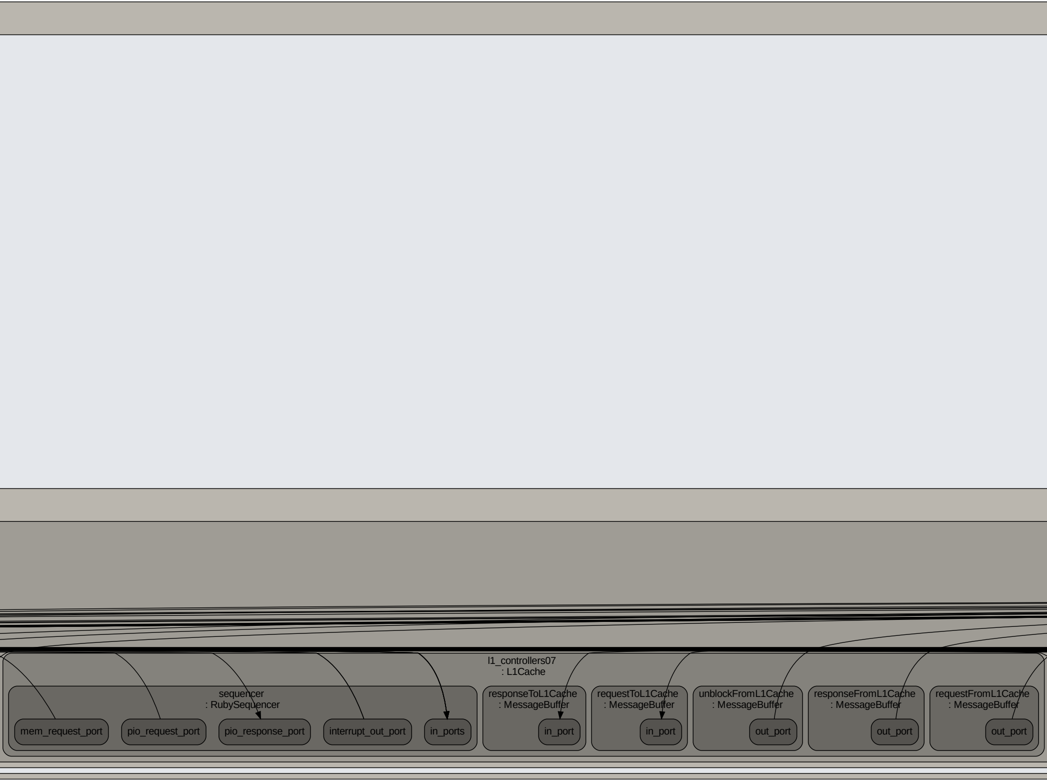

root
: Root
board
: X86Board
l1_controllers12
: L1Cache
unblockFromL1Cache
: MessageBuffer
out_port
responseFromL1Cache
: MessageBuffer
out_port
requestFromL1Cache
: MessageBuffer
out_port
sequencer
: RubySequencer
mem_request_port
pio_request_port
pio_response_port
interrupt_out_port
in_ports
responseToL1Cache
: MessageBuffer
in_port
requestToL1Cache
: MessageBuffer
in_port
unblockFromL1Cache
: MessageBuffer
out_port

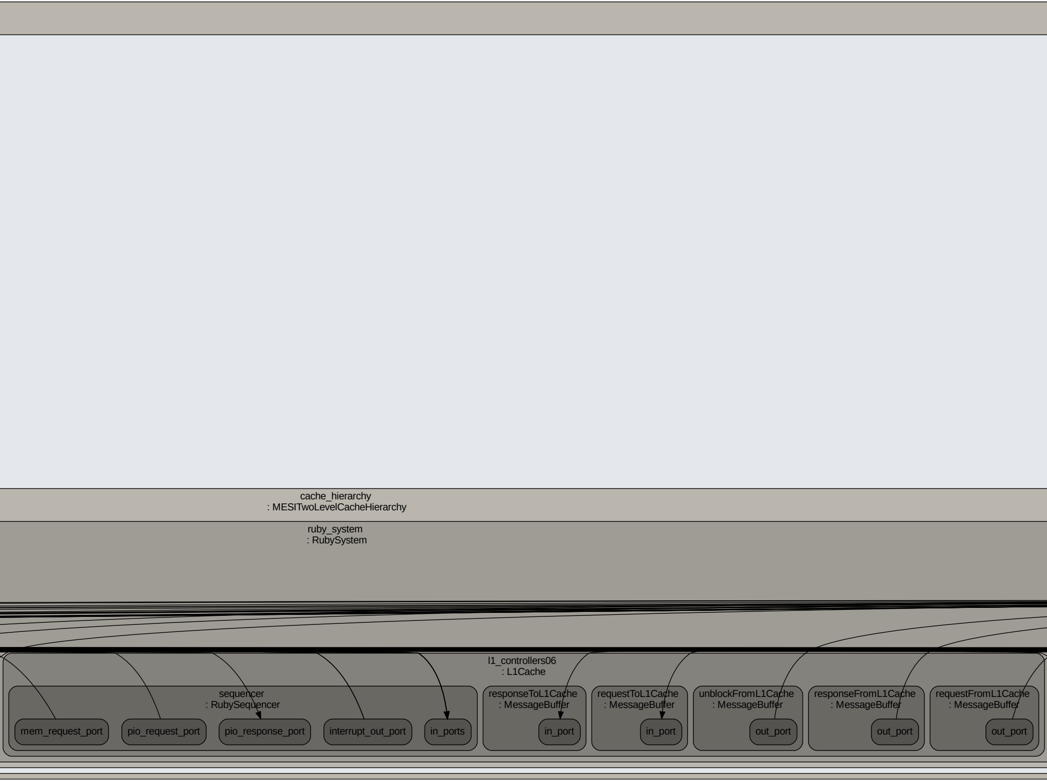

l1_controllers11
: L1Cache
responseFromL1Cache
: MessageBuffer
out_port
requestFromL1Cache
: MessageBuffer
out_port
sequencer
: RubySequencer
mem_request_port
pio_request_port
pio_response_port
interrupt_out_port
in_ports
responseToL1Cache
: MessageBuffer
in_port
requestToL1Cache
: MessageBuffer
in_port
unblockFromL1Cache
: MessageBuffer
out_port
responseFromL1Cache
: MessageBuffer
out_port

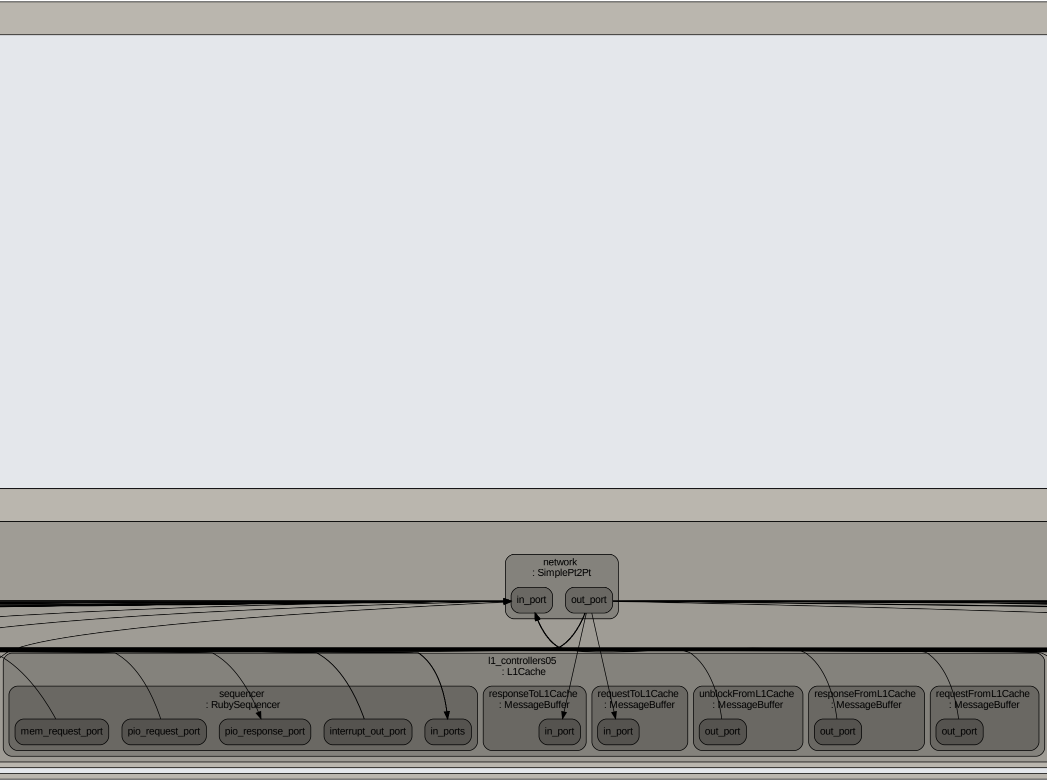

system_port
sys_port_proxy
: RubyPortProxy
in_ports
l1_controllers10
: L1Cache
requestFromL1Cache
: MessageBuffer
out_port
sequencer
: RubySequencer
mem_request_port
pio_request_port
pio_response_port
interrupt_out_port
in_ports
responseToL1Cache
: MessageBuffer
in_port
requestToL1Cache
: MessageBuffer
in_port
unblockFromL1Cache
: MessageBuffer
out_port
responseFromL1Cache
: MessageBuffer
out_port
requestFromL1Cache
: MessageBuffer
out_port

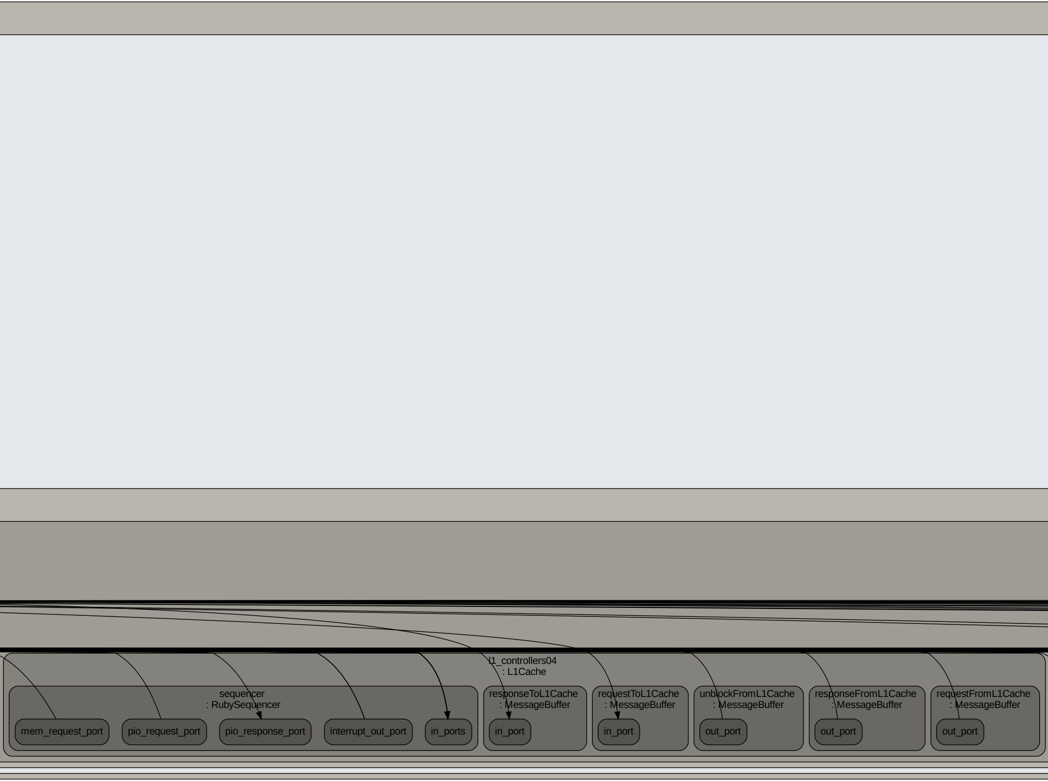

l1_controllers09
: L1Cache
sequencer
: RubySequencer
mem_request_port
pio_request_port
pio_response_port
interrupt_out_port
in_ports
responseToL1Cache
: MessageBuffer
in_port
requestToL1Cache
: MessageBuffer
in_port
unblockFromL1Cache
: MessageBuffer
out_port
responseFromL1Cache
: MessageBuffer
out_port
requestFromL1Cache
: MessageBuffer
out_port

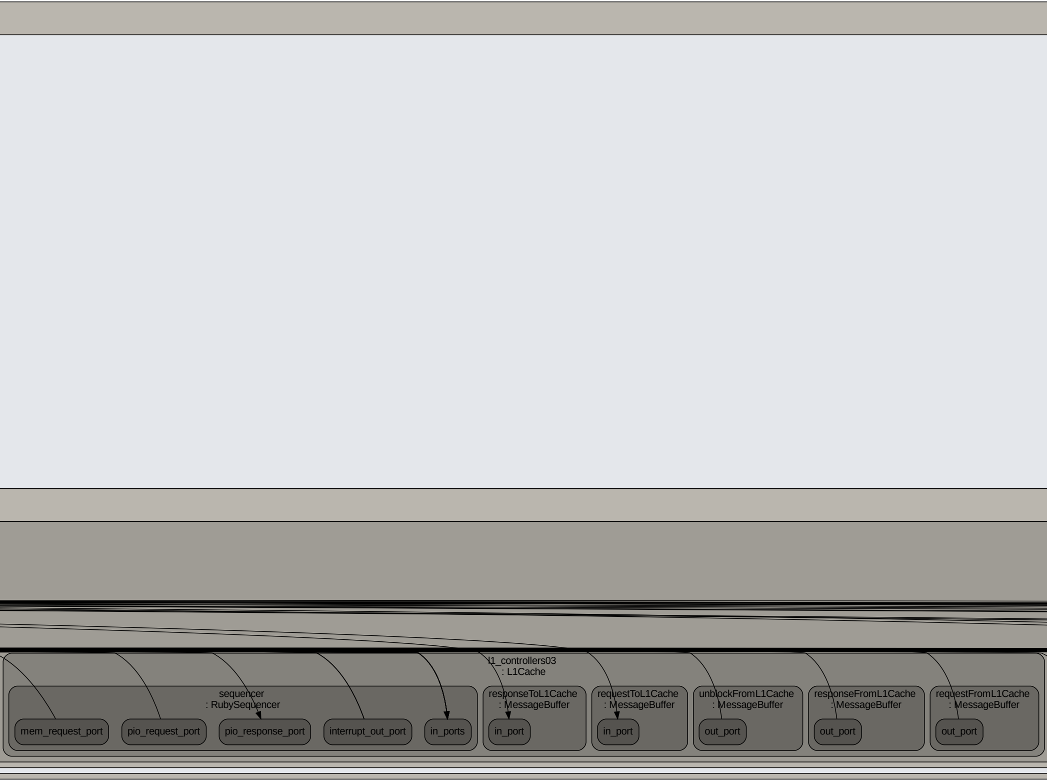

l1_controllers08
: L1Cache
sequencer
: RubySequencer
mem_request_port
pio_request_port
pio_response_port
interrupt_out_port
in_ports
responseToL1Cache
: MessageBuffer
in_port
requestToL1Cache
: MessageBuffer
in_port
unblockFromL1Cache
: MessageBuffer
out_port
responseFromL1Cache
: MessageBuffer
out_port
requestFromL1Cache
: MessageBuffer
out_port

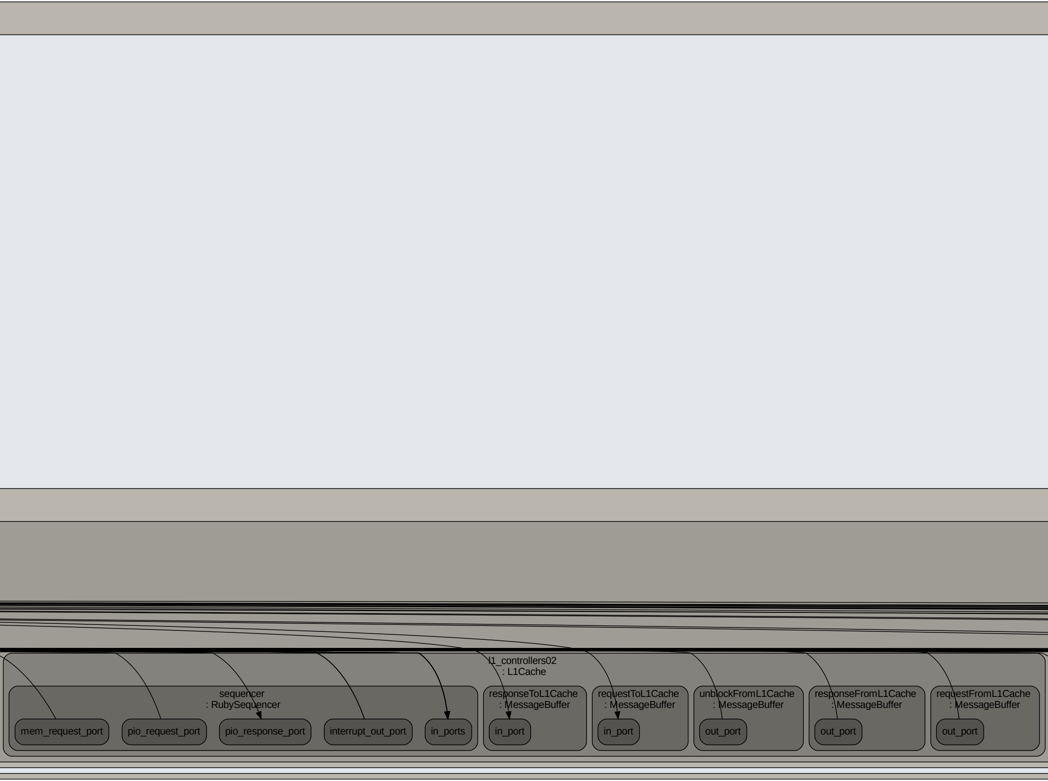

l1_controllers07
: L1Cache
sequencer
: RubySequencer
mem_request_port
pio_request_port
pio_response_port
interrupt_out_port
in_ports
responseToL1Cache
: MessageBuffer
in_port
requestToL1Cache
: MessageBuffer
in_port
unblockFromL1Cache
: MessageBuffer
out_port
responseFromL1Cache
: MessageBuffer
out_port
requestFromL1Cache
: MessageBuffer
out_port

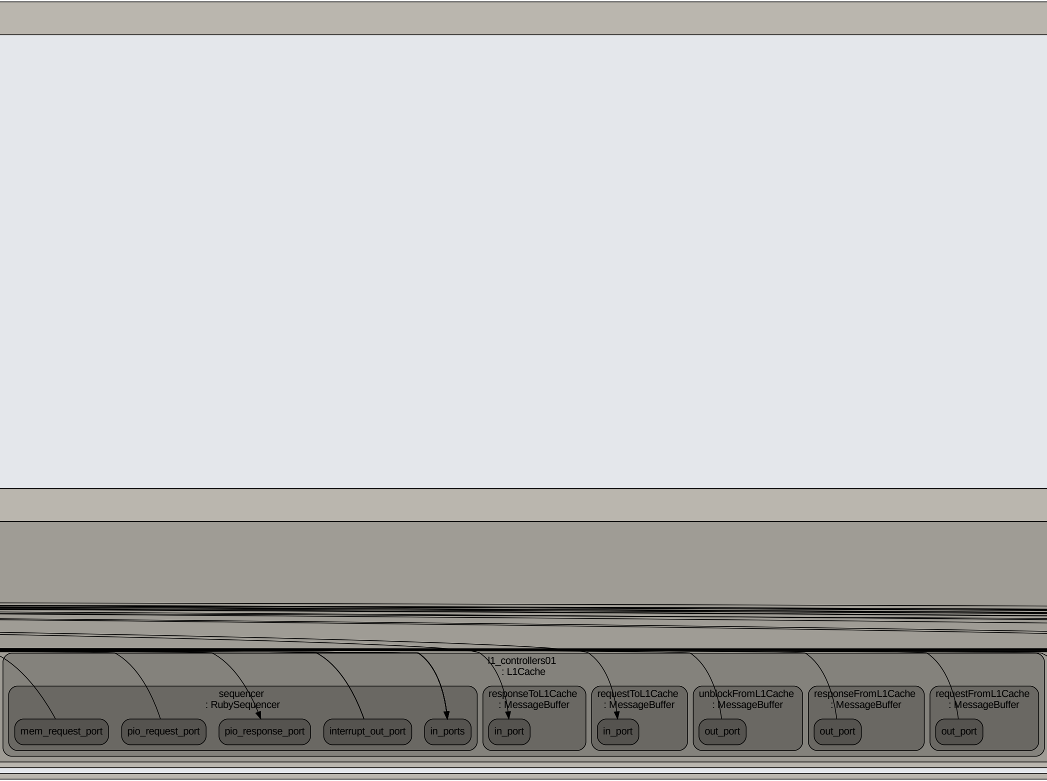

cache_hierarchy
: MESITwoLevelCacheHierarchy
ruby_system
: RubySystem
l1_controllers06
: L1Cache
sequencer
: RubySequencer
mem_request_port
pio_request_port
pio_response_port
interrupt_out_port
in_ports
responseToL1Cache
: MessageBuffer
in_port
requestToL1Cache
: MessageBuffer
in_port
unblockFromL1Cache
: MessageBuffer
out_port
responseFromL1Cache
: MessageBuffer
out_port
requestFromL1Cache
: MessageBuffer
out_port

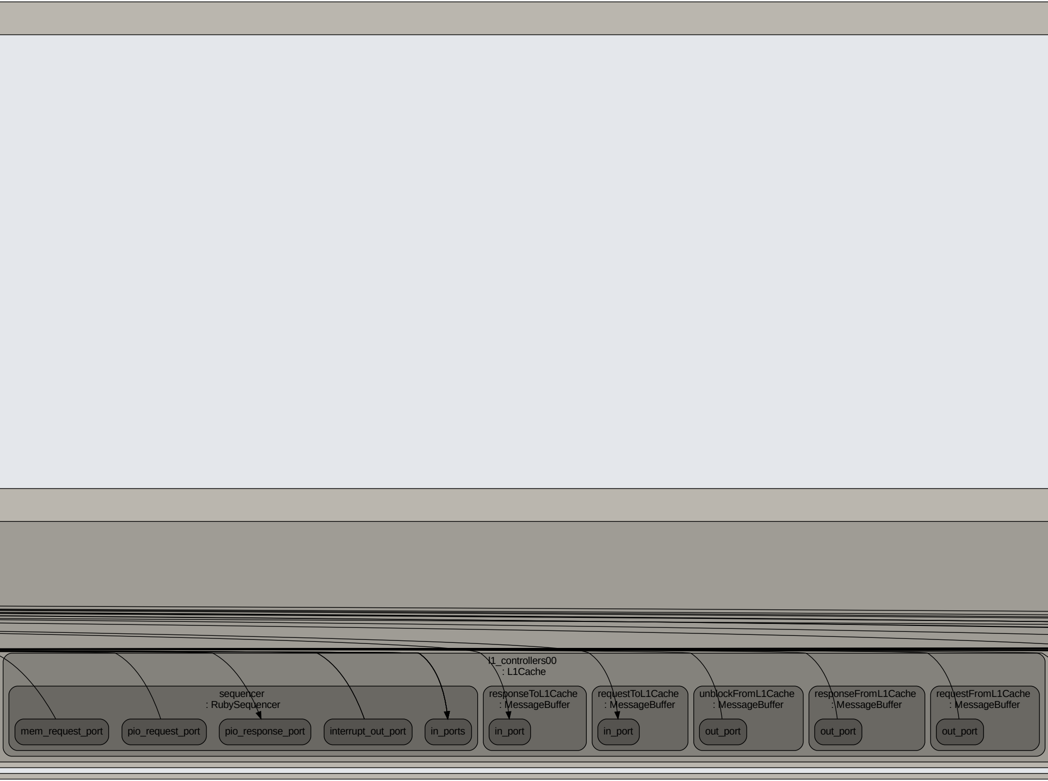

network
: SimplePt2Pt
in_port
out_port
l1_controllers05
: L1Cache
sequencer
: RubySequencer
mem_request_port
pio_request_port
pio_response_port
interrupt_out_port
in_ports
responseToL1Cache
: MessageBuffer
in_port
requestToL1Cache
: MessageBuffer
in_port
unblockFromL1Cache
: MessageBuffer
out_port
responseFromL1Cache
: MessageBuffer
out_port
requestFromL1Cache
: MessageBuffer
out_port

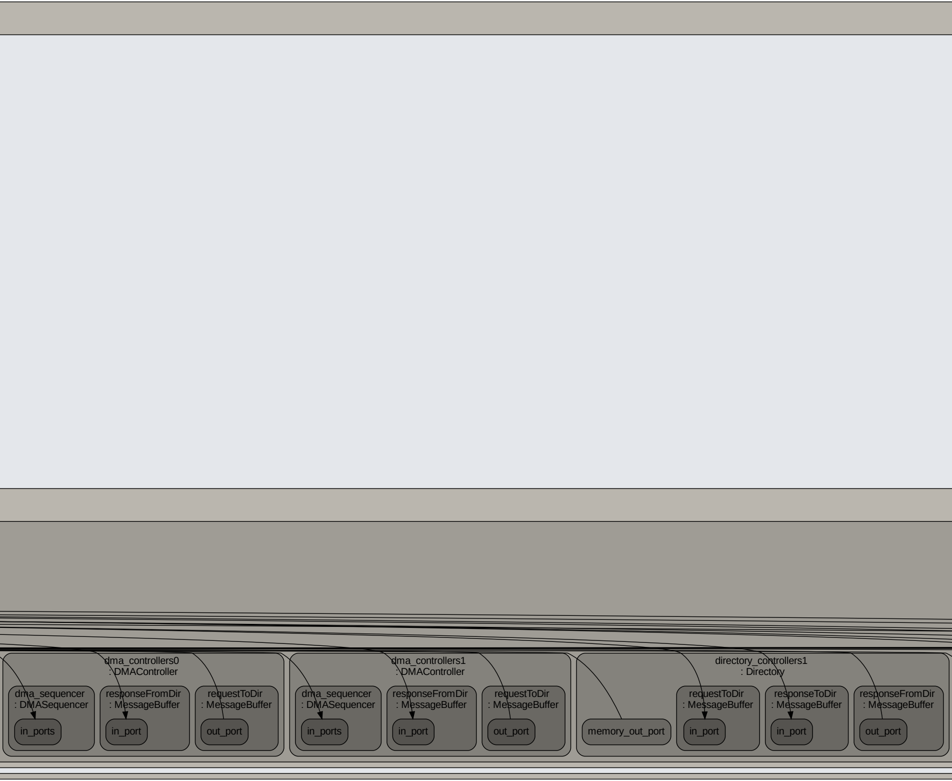

l1_controllers04
: L1Cache
sequencer
: RubySequencer
mem_request_port
pio_request_port
pio_response_port
interrupt_out_port
in_ports
responseToL1Cache
: MessageBuffer
in_port
requestToL1Cache
: MessageBuffer
in_port
unblockFromL1Cache
: MessageBuffer
out_port
responseFromL1Cache
: MessageBuffer
out_port
requestFromL1Cache
: MessageBuffer
out_port

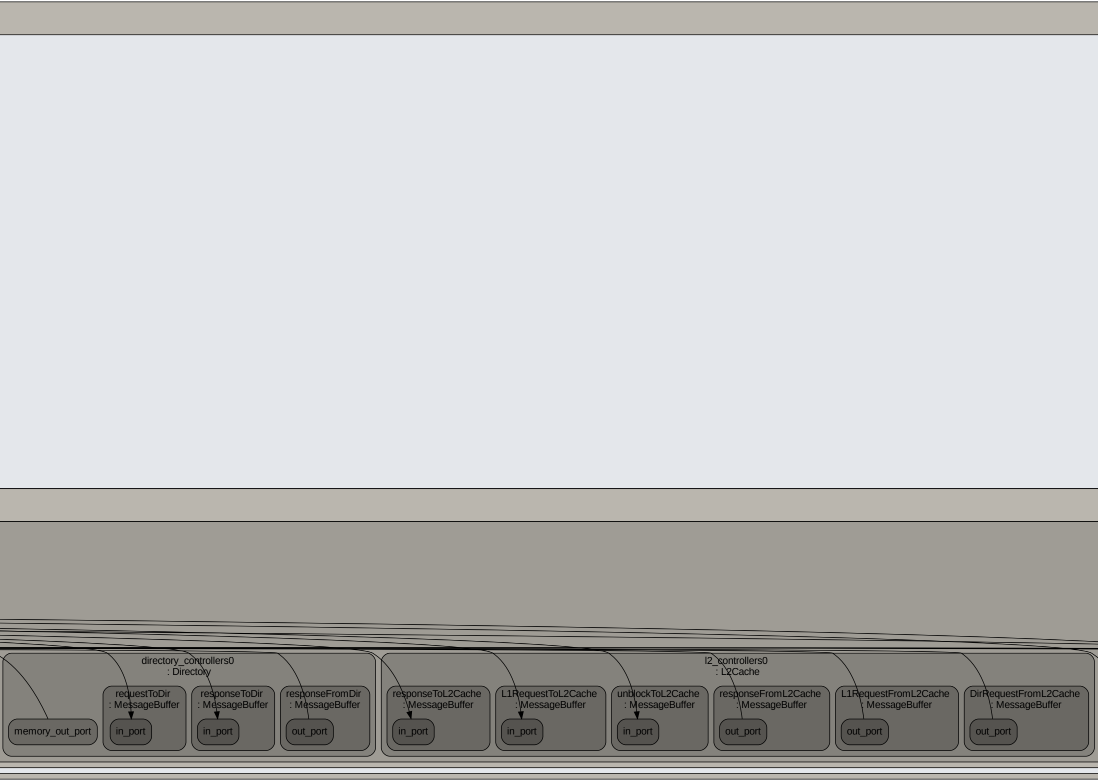

l1_controllers03
: L1Cache
sequencer
: RubySequencer
mem_request_port
pio_request_port
pio_response_port
interrupt_out_port
in_ports
responseToL1Cache
: MessageBuffer
in_port
requestToL1Cache
: MessageBuffer
in_port
unblockFromL1Cache
: MessageBuffer
out_port
responseFromL1Cache
: MessageBuffer
out_port
requestFromL1Cache
: MessageBuffer
out_port

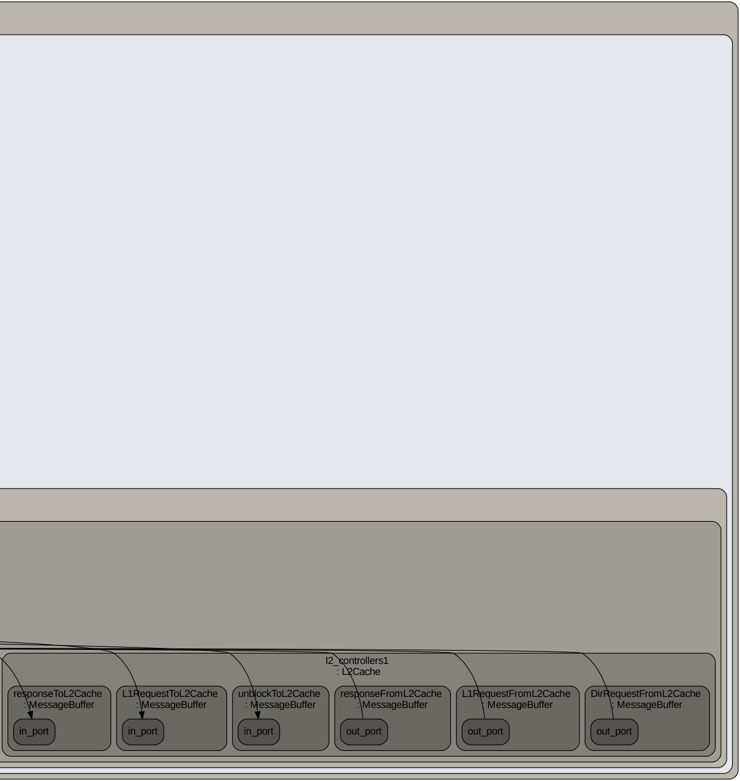

l1_controllers02
: L1Cache
sequencer
: RubySequencer
mem_request_port
pio_request_port
pio_response_port
interrupt_out_port
in_ports
responseToL1Cache
: MessageBuffer
in_port
requestToL1Cache
: MessageBuffer
in_port
unblockFromL1Cache
: MessageBuffer
out_port
responseFromL1Cache
: MessageBuffer
out_port
requestFromL1Cache
: MessageBuffer
out_port


l1_controllers01
: L1Cache
sequencer
: RubySequencer
mem_request_port
pio_request_port
pio_response_port
interrupt_out_port
in_ports
responseToL1Cache
: MessageBuffer
in_port
requestToL1Cache
: MessageBuffer
in_port
unblockFromL1Cache
: MessageBuffer
out_port
responseFromL1Cache
: MessageBuffer
out_port
requestFromL1Cache
: MessageBuffer
out_port


l1_controllers00
: L1Cache
sequencer
: RubySequencer
mem_request_port
pio_request_port
pio_response_port
interrupt_out_port
in_ports
responseToL1Cache
: MessageBuffer
in_port
requestToL1Cache
: MessageBuffer
in_port
unblockFromL1Cache
: MessageBuffer
out_port
responseFromL1Cache
: MessageBuffer
out_port
requestFromL1Cache
: MessageBuffer
out_port


dma_controllers0
: DMAController
dma_sequencer
: DMASequencer
in_ports
responseFromDir
: MessageBuffer
in_port
requestToDir
: MessageBuffer
out_port
dma_controllers1
: DMAController
dma_sequencer
: DMASequencer
in_ports
responseFromDir
: MessageBuffer
in_port
requestToDir
: MessageBuffer
out_port
directory_controllers1
: Directory
memory_out_port
requestToDir
: MessageBuffer
in_port
responseToDir
: MessageBuffer
in_port
responseFromDir
: MessageBuffer
out_port


directory_controllers0
: Directory
memory_out_port
requestToDir
: MessageBuffer
in_port
responseToDir
: MessageBuffer
in_port
responseFromDir
: MessageBuffer
out_port
l2_controllers0
: L2Cache
responseToL2Cache
: MessageBuffer
in_port
L1RequestToL2Cache
: MessageBuffer
in_port
unblockToL2Cache
: MessageBuffer
in_port
responseFromL2Cache
: MessageBuffer
out_port
L1RequestFromL2Cache
: MessageBuffer
out_port
DirRequestFromL2Cache
: MessageBuffer
out_port


l2_controllers1
: L2Cache
responseToL2Cache
: MessageBuffer
in_port
L1RequestToL2Cache
: MessageBuffer
in_port
unblockToL2Cache
: MessageBuffer
in_port
responseFromL2Cache
: MessageBuffer
out_port
L1RequestFromL2Cache
: MessageBuffer
out_port
DirRequestFromL2Cache
: MessageBuffer
out_port